Assessed Coursework Cover Sheet for PSYC0098

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INTRODUCTION

This report presents a comprehensive reflection of two distinct activities conducted in the course “Human Factors for Digital Health”, aiming to develop a deep understanding of behaviour change theories and intervention techniques with a practical backdrop. The report consists of two parts: the first focuses on reflecting on the experience of evaluating a digital behaviour change intervention (DBCI) through auto-ethnography, and the second discusses and reflects on a group activity of health management software prototyping and evaluation.

By examining, discussing and reflecting on the strategies used and challenges faced throughout the course, this report provides a deeper understanding of effective individual work and teamwork in addressing complex behaviour and health issues. in the realm of digital health research.

PART 1: AUTO-ETHNOGRAPHY

Aim

The aim of this study is to record and study the experience of using a habit-forming App HABITICA to better understand Digital Behavioural Change Intervention (DBCI) theories and obtain insights for the design of behaviour change technologies.

Background

Upon moving to London for my master's degree, I became aspired to start a company in three years. However, I recognized a prominent need to improve various aspects of my life, namely sleep, focus, social skills, and physicality in order to realise my dream. Before starting to use HABITICA I was already using a paper-based checklist for five habits: meditation, exercise, socialization, reading, and journaling, and had moderate results. I aim to gradually consolidate all of them so I can be healthier and more energetic every day. Driven by academic need and personal curiosity (I was sceptical about the effectiveness of gamification), I chose HABITICA, a gamified habit-forming App as my DBCI, to see if it can help me achieve my goal: consolidate my 5 daily habits.

Strategy, Activities and Decisions

Throughout the course of 3 weeks, I wrote about my thoughts on my experience of using HABITICA in my diary. In total, 11 meaningful entries were made. Although no rigorous structure was applied when writing my diary, 4 questions about my experience were frequently answered:

- What went well that day?
- What didn’t go so well that day?
- What role this App is playing in my day?
- What is my current mood? (this is answered implicitly)

Furthermore, there were also a few days I reflected on my behaviour change extensively with regard to behaviour change and design theories, which is worth noting.

Two extracts are presented in the appendix as an example.

Thematic analysis was conducted to analyse the auto-ethnography. Key activities during data analysis include Code Extraction, Theme Generation and Theory Reflection.

- Code Extraction: During the first round of reading through my diary, I hand-coded subjectively meaningful content with the aim to identify my attitude, behaviour and comments on App features. The bottom-up approach was used.
- Theme Generation: After code extraction, they were first arranged by the tone of quotes (negative, positive or neutral) and then semantically grouped into higher-level code in an affinity diagram. Several emerging themes were identified.
- Reflection: Lastly, the themes were revised by re-reading the diary and reflecting on my experience through memory with the help of several behaviour change models, including Decision Theory and Reasoned Action Approach.

During Reflection, some trivial codes were eliminated due to low relevance to behaviour change, while a few more codes were added in forming the theme “Coping with Obstacles” as they were initially overlooked, and later found out to be highly relevant to self-maintenance.

Result

8 higher level codes were identified from the affinity diagram, which was eventually grouped into 3 key themes: Habit Consolidation, Personal Growth and Limitations.

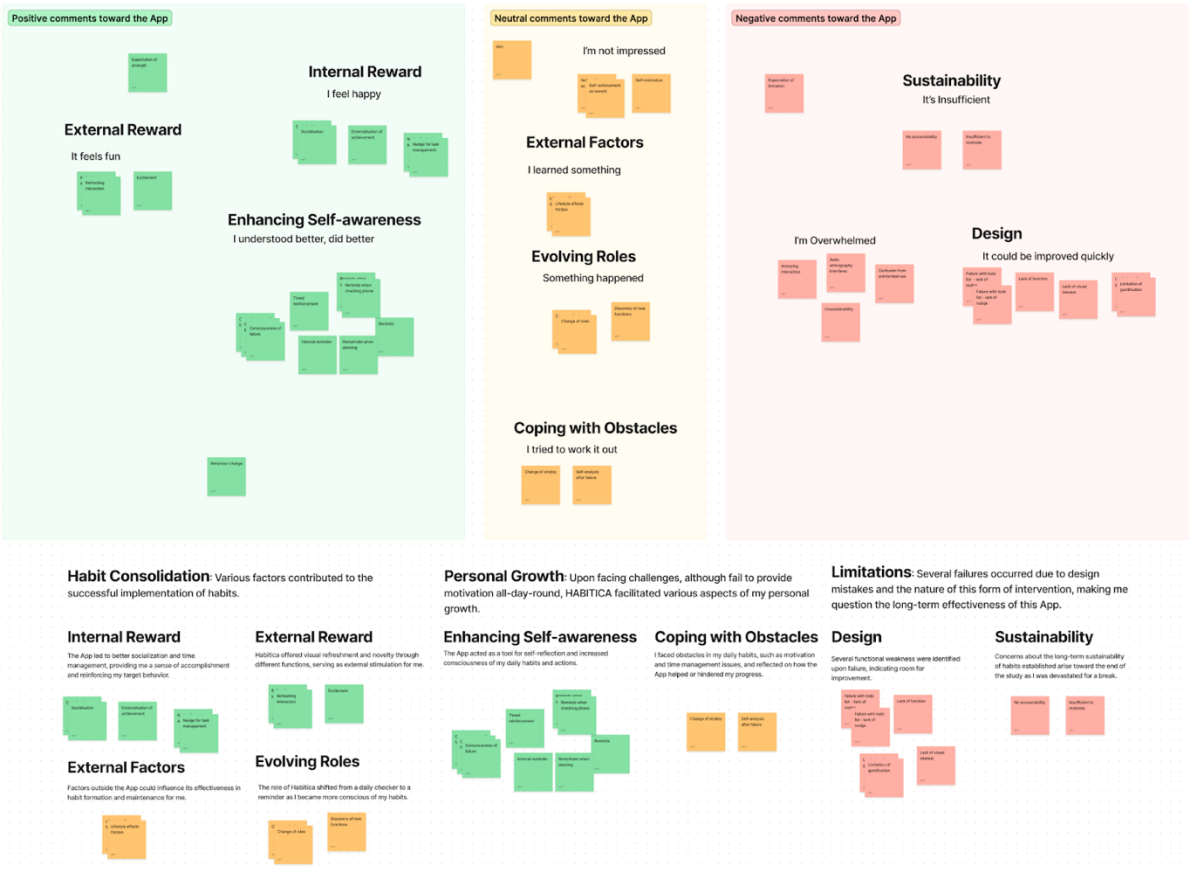

Figure 1. Affinity Diagram and key themes

Habit Consolidation: HABITICA has brought me both internal and external rewards to reinforce my habits. Because I can open HABITICA on my phone anytime to track my progress, I felt a better sense of achievement. The gamification became a good conversation starter to happily share self-improvement achievements with people. They are both internal rewards (from myself). The visually refreshing animations also boosted my mood during the first week of use, acting as an external reward. External factors affect the effectiveness of these rewards, such as sudden changes in lifestyle. The role of the App had evolved from being an easily accessible checkbox to a reminder, as my performance progressed. At the end of the experiment, 3 habits (journaling, exercising, and meditation) were consolidated.

Personal Growth: There are a couple of days where I failed to perform certain habits, but because of the existence of HABITICA (and possibly the duty of auto-ethnography), I managed to reflect and enhance my self-awareness regarding my performance and came up with strategies to deal with obstacles.

Limitations: Upon failure to perform certain habits, functional deficiencies were identified immediately, such as lack of visual interest and lack of nudging. In the end, although 3 habits were consolidated, I still questioned the sustainability of this consolidation, as I became frustrated about the other two failures. I criticised the different colour coding for habits that reminds you that “socialising and reading are lagging behind others” gave me a sense of defeat and concluded that I there is no point to carry on as they will never align from now.

Discussion

Autoethnography as a method to evaluate DBCI offers valuable insights into personal experiences and emotions while other types of evaluation methods, such as randomised control trials cannot. It is especially effective at capturing the intricacies of the user's journey such as attitudes and mood. However, limitations exist. For example, in my diary, I mentioned explicitly that the duty to write auto-ethnography affects my attitude and behaviour – a confounding factor. Furthermore, as individuals' ability to accurately recall and interpret their experiences strongly affects the data, the outcome is prone to low generalisability.

If I were to conduct another autoethnography, I would consider bringing more structure into entries such as setting questions for me to answer, instead of writing in free form. This would provide me with more comprehensive and organised data, reducing the burden of analysis. However, I would maintain my reflective diary approach, as it was efficient in capturing my thoughts, emotions and ideas. Therefore, I would conduct a semi-structured reflective diary next time.

Reflection

Throughout my autoethnography journey, I learned how HABITICA, as a Digital Behaviour Change Intervention (DBCI), influenced my habits, particularly in relation to the Reasoned Action Approach (RAA) [2] COM-B [4] and Decision Theory [7].

Considering RAA, by acting as a conversation starter about my self-improvement achievements, the App implicitly boosted my self-efficacy [2]. However, it failed to

account for changing external and internal factors, such as opportunity and motivation [4] (from the COM-B model). In the middle of the experiment, when I was extremely busy and did not have the opportunity to socialize and read, my self-efficacy dropped significantly. Nevertheless, my intention was firm, so I adopted a new strategy to socialise and succeeded for 2 days. From RAA's closed-loop perspective, the drop in self-efficacy comes from the change of internal factors, and it will lead to a further drop in performance (relapse of behaviour change) down the line, which was not the case in my experience.

As for Decision Theory, at the beginning of the experiment, both external and internal rewards were built by the App, but the immediate cost [7] was not imminent. In the end, the cost of not performing built up unnoticed through the counting of each checkbox and eventually emerged as unaligned colours (indicating 'level of achievement') for checkboxes, leading to surprise, frustration and abandonment. Therefore, from the Decision Theory perspective, HABITICA fails to sustain long-term habit formation.

PART 2: DESIGN PROJECT

Aim

The aim of this project is to address the symptoms of long-term COVID-19 patients. We settle our focus on sleep disturbances as it is being increasingly prevalent and could lead to further problems that can severely impact life quality, such as fatigue, poor concentration, and poor mental health [9][10]. Therefore, our primary goal was to create a digital intervention in the form of a mobile App to promote bedtime relaxation, quicken sleep onset, and eventually improve sleep quality, hoping to eventually help return to their pre-COVID state of functioning.

Strategy, Challenges and Decisions

We employed the Double Diamond decision-making strategy during user research. During the user research stage, we first conducted literature reviews to obtain basic demographic and epidemiologic information about Long-COVID and narrowed our focus to sleep disorders. We then conducted semi-structured interviews to dig out deeper insights. Lastly, we generated a series of personas presenting key demographics of our user base and turned them into design scenarios.

The Iterative Development strategy was used during prototype design. We started our prototype design with a group brainstorm and landed on the concept of a gamified sleep assistance mobile App through discussion and voting. Then we collaboratively developed the low-fidelity prototype for user testing. In the end, we created a hi-fi prototype based on the user experience insights collected from testing.

Despite our structured approach, we encountered a few challenges during this group project and made several critical decisions to push forward the progress. Due to the group size (group of nine) and diverse backgrounds of members, we struggled to narrow down our focus after the literature review and spend a significant amount of time in inefficient communication. Therefore, we decided to split into 3 subgroups to conduct interviews and generate personas. This decision resulted in higher efficiency but also caused stagnation during the general discussion late.

After design idea brainstorming, we selected the best idea for our App through subjective voting. However, we realised that the winning concept “building a community for stress release and reducing feelings of loneliness” lacks a firm definition of the target behaviour, as well as many other idea definitions. Therefore, we decided to rewrite our idea definition into a descriptive sentence of behaviour change goal and vote again, this time with the perspective of the COM-B model. In the end, we selected our final design idea “Practice mindfulness before bed with the help of a gamified sleep assistance App”.

The end product we delivered was a sleep companion App called SleePal. Apart from the DBCI models we applied during early development, we have also adopted a user-centred design approach in later design stages by applying usability heuristics [5] to further improve the interaction flow of our high-fidelity prototype. One of the key features of this App is that it is centred around a cute virtual figure called Dingo, and a narrative was built around it. We introduced this figure in the hope to enhance user relatedness, referring to the Self-determination Theory [1]. Although the effect may vary between individuals due to the nature of virtual figures.

Discussion

The double diamond model provided us with a firm structure to follow during the fuzzy stage of topic research and nudged us to make decisions based on the stage we were at (diverge research and converge decisions). The model’s deep root in design also meant that the engagement of students with a design background (such as me) was ensured during the research stage.

Limitations to this model do exist, although the aforementioned splitting up temporarily improved efficiency, it also prevented group members to get to know each other in this critical stage, resulting in reluctance in raising opinions during the general discussion that followed, eventually slowing down the converging process. The same problem was observed during the voting stage after idea brainstorming. Although this problem cannot be attributed solely to the double diamond strategy itself, it does indicate the complicated nature of group work management and the insufficiency of only using one decision model.

Looking back at this stage of group work in hindsight, the diverse background of our group did not contribute much to our progression but rather slowed down many decision-making processes. However, in the end, it resulted in a rich database of user information and has benefited us in the long run. Given more time to conduct this project, I believe a diverse group can be more advantageous.

The iterative design approach was loosely followed during the prototyping stage, which was reflected by the two stages of prototyping and the user testing activity conducted between them. Reflecting on the experience, an apparent strength of this approach is that key user interaction errors were spotted. By reading through the user feedbacks together, we developed a more rigorous mental framework of user needs and pain points, hence the decision process for collaborative hi-fi prototyping was smoother.

The main drawback of this approach was time-consuming. During the last few weeks, we decided to not have general meetings weekly and readopted the split-up strategy, where HCI (Human-computer Interaction) and BC (Behaviour Change) students were in charge of App design and evaluation, and DHE (Digital Health and Entrepreneurship) students were in charge of recording the whole process and report write-up. This was because the amount of feedback we obtained from user testing was disproportionate to the time left for redesign, and we faced severe time management obstacles. If we were not conducting this round of evaluation, we could have had the opportunity to discuss the implementation of DBCI models across the group, and potentially produce better work.

A few other challenges we faced during this stage are worth noting. Because design protocols such as Crazy 8 Sketching and Decision Matrix were foreign to students from outside HCI, design collaboration was initially troublesome. Their unfamiliarity with design software such as Figma adds even more burden to collaboration. Luckily, I and another teammate from HCI eventually managed to take the lead and facilitated our collaborative design sessions. On the other hand, HCI students have struggled in understanding the application of DBCI models proposed by our BC member. For example, when explaining how the reflective and automatic motivation from the COM-B model defers in the case of Cathy (one of the personas), our HCI members (including me) were confused by the specialised terms and phrases used by BC students. Nevertheless, we eventually overcame these challenges and successfully delivered our design.

Reflection

Given another opportunity to conduct such a study, I would retain the two decision strategies on position, but try to bring a more vibrant structure into our group work and introduce more group design techniques to increase efficiency.

The double-diamond approach has provided us with a solid framework to refer to constantly, and to a certain degree it has alleviated our mental workload when doing research, especially for students from DH. This was because the concept of convergence is straightforward to apply, and served as a reminder for us to narrow down our scope at appropriate time points.

The iterative design approach, specifically conducting think-aloud studies after low-fidelity prototyping in the wild, has offered us rich data to improve the usability of our App. We would also struggle to apply usability heuristics to improve our high-fidelity prototype if there were no real-world comments to refer to. The Crazy 8 Sketching and the Subjective Voting technique in the early stage of collaborative design have also contributed to a more comprehensive initial concept and sped up the prototype design process.

Despite the convenience of only referring to a single, well-established decision model in each stage, we observed some groupthink [3] effects here and there. For example, when we tried to reduce the number of personas to more generalisable user scenarios, stagnation happened due to the lack of rapport among team members. Similarly, during voting for design ideas, we overlooked the importance of applying DBCI models out of the desire to maintain the harmony of discussion, leading to another round of voting.

Although it is hard to counteract groupthink effects with individual effort, I still believe by implementing the strategies next time, our group work efficiency can be increased.

Firstly, I would aim to establish a more rigorous guideline for decision-making, as suggested by Tjosvold [8]. When selecting wireframe concepts, we used the decision matrix to guide our selection, and it was proven to be efficient. Next time, I shall introduce this technique earlier to the group, preferably every time stagnation happens, so that we can reduce the reluctance to raise opposing opinions and speed up our decision-making progress.

Secondly, during voting sessions, I would suggest anonymising both the votes and the follow-up comments. To anonymise the follow-up feedback, we would need to adopt a next-based discussion in a specifically prepared chat room. This process would be helpful as it would allow members to express critical opinions towards each other more freely [6], and avoid the formation of an echo chamber.

Apart from the groupthink effects, time management and knowledge gap are two other problems I should take care of next time. For time management, I would propose breaking down our tasks into smaller pieces by setting more deadlines for each stage of the project. This will reduce the perceived workload for our group members and increase the sense of achievement after each session. For the knowledge gap between students with different backgrounds, I would organise experience orientate workshops early in the project, to get all of our members familiarised with important concepts and skills of each discipline. This would be more efficient than distributing learning sessions across project meetings as the latter has been shown in our experience to put massive pressure on our members.

In conclusion, reflecting on my experience, I recognise the complex nature of group work management and the challenges I could still face in conducting group projects in the future. I believe by carefully applying decision frameworks, strategies and specific group work techniques, me and my group can better handle similar projects in the future. Maximising the effectiveness of communication, decision-making, and collaboration. In this way, we as a multidisciplinary team will have a solid foundation to collaborate and deliver more successful digital behaviour intervention technologies.

SUMMARY

In this reflective report, I have presented and reflected on my experience of conducting an individual auto-ethnographical review of a digital behaviour change intervention (DBCI) mobile application and a group project that delivered a health management software prototype that targeted sleep disorder caused by Long-COVID, with a focus of understanding and applying behaviour change theories.

In the individual study, I discovered that auto-ethnography is advantageous when looking for personal and in-depth data, but could also cause severe bias. I have also identified the strength and weaknesses of different behaviour change models

when evaluating DBCI. My personal experience of recording auto-ethnography has highlighted the need to carefully design this approach to reduce confounding factors.

From the group project, I realised the importance of applying behaviour change models early and comprehensively in order for an effective design outcome. Additionally, the complex nature of group work emphasises a strong need for a rigorous decision framework and careful facilitation. Reflecting on my personal role, I have identified some missed opportunities for me as a designer to use various techniques to increase our work efficiency.

In my future career, if I were to conduct another diary study (auto-ethnography in design context) for technology evaluation, I will develop a structure and set of requirements as a semi-formal instruction for my entry, as I believe this could reduce my mental and physical when of writing the diary, hence maintain a higher level of isolation for the measured activity/experience. When conducting group research/design, I will strive to enhance my role as an HCI designer that foresees potential communication struggles and mitigate them with various design and psychological strategies. Furthermore, I will also step in promptly to manage knowledge and skill gaps when working in a multidisciplinary setting.

Overall, reflecting on my experience throughout the course has consolidated my theoretical base as an HCI designer and provided me with valuable practical insights, enabling me to better contribute to more impactful design and research projects in the future.

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